Solar Energy Factsheet

How solar panels work

Solar panels use the photovoltaic effect to turn sunlight into direct current (DC) electricity. An inverter then converts the DC electricity into alternating current (AC), which is used by household appliances. Most home panels today are made of silicon cells.

Panel types

Monocrystalline panels have efficiencies between 20 and 22 percent and cost more. Polycrystalline panels have efficiencies between 15 and 17 percent and are cheaper. Thin-film panels are lightweight but have lower efficiency, around 10 to 13 percent.

Performance and maintenance

Solar panels lose about 0.5 percent of their output each year. Most manufacturers give a performance warranty of 25 years. The typical payback time for a residential system is between 6 and 10 years.

Panels should be cleaned about twice a year, and the best tilt angle is roughly equal to the latitude of the installation site.

Battery storage

A battery storage system lets households use solar energy at night. A typical home battery stores between 5 and 15 kWh.